Instructions to access the spmcluster.unipi.it cluster

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To access the cluster infrastructure, connect via SSH to the following machine:

spmcluster.unipi.it (IP: 131.114.51.114)

The frontend node provides access to a 10-node cluster (one frontend and nine compute nodes). The first 8 nodes (node01-node08) are homogeneous (same microarchitecture), whereas the nineth node (node09) has a different architecture (AMD-based) and is equipped with NVIDIA A30 GPU.

To gain access, please email me your **public SSH key** with the subject line "**SPM SSH key**". Below are the instructions for generating your SSH key pair:

For Unix-based machines

- Generate the SSH key using the command: ssh-keygen -t ed25519 (or -t rsa if ed25519 is unavailable)
- When prompted, choose where to save the key. Press Enter to accept the default (e.g., ~/.ssh/id_ed25519).
- You may enter a passphrase (optional).
- Unless you chose a different name, your public SSH key is ~/.ssh/id_ed25519.pub (or id_rsa.pub).

For Windows-based machines

- Option1: Install WSL2 (Windows Subsystem for Linux) on your machine and then install the Ubuntu distribution (or any other Linux distro you prefer). At the end of the process, you will have a Linux distribution running under Windows, allowing you to use the previous Unix-based machine commands to generate the keys.
  Instructions for installing WSL: https://learn.microsoft.com/en-us/windows/wsl/install
  Basic WSL commands: https://learn.microsoft.com/en-us/windows/wsl/basic-commands
- Option2: Install OpenSSH for Windows and generate a key pair as described in the Microsoft's guide. Instructions for installing OpenSSH and generating the key pair can be found here: https://learn.microsoft.com/en-us/windows-server/administration/openssh/openssh_install_firstuse?tabs=gui

Account creation

After I receive your public key via email, I will set up your account on the frontend node of the cluster. This may take a few days. Please note that you **will not receive a confirmation email, so please try logging in after some time** to check if your account has been set up.

VPN (off-campus only)

To access the cluster via SSH **from outside the unipi.it domain**, you **must** first connect to the UNIPI Virtual Private Network (VPN).

The instructions for setting up the UNIPI VPN can be found here:

https://it.unipi.it/configurazioni/sicurezza-e-privacy/vpn-di-ateneo/

https://www.sba.unipi.it/en/services/how-access-e-resources

Pay attention to installing "Connect Tunnel" (you have to scroll down the page and find **Connect Tunnel**) and activating the VPN by accessing the server "**access.unipi.it**". Use your UNIPI credentials, which are the same as those for Alice, email, and all other UNIPI services.

IMPORTANT: Ensure that you have activated the VPN UNIPI service in the following portal

https://autenticazione.unipi.it/auth/auth.signin

SSH login

You can access the frontend node by using this SSH command:

    ssh LOGINNAME@spmcluster.unipi.it

LOGINNAME is the username part of your email address (e.g., for the student Mario Rossi, who has the email address **m.rossi21**@studenti.unipi.it, the username is **m.rossi21**). Note that everything should be lowercase. No password is required unless you set a passphrase when generating your SSH key.

File transfer

To transfer data to or from the machines, you can use programs like scp, rsync, sftp, or FileZilla. For example, using rsync:

- rsync -av tocopy loginname@spmcluster.unipi.it:~/dir/
  - it copies the tocopy file/dir into the remote machine's directory ~/dir/)
- rsync -av loginname@spmcluster.unipi.it:~/myTests/test1 ./
  - It copies the test1 file/dir into the local machine's current directory (./).

Watch out for the final '/' in rsync commands

Be careful with the destination path when using rsync, especially with the -a flag. If you copy files directly to ~/ (home directory) instead of to a subdirectory, rsync may change the permissions of your home directory. On the cluster, home directories must have permissions 700. If permissions become more permissive, for example, 755 or 775, SSH will refuse the connection, and you will not be able to log in anymore.

Recommended usage:
OK, copy into a subdirectory
rsync -rv data/ username@spmcluster:~/data/

To avoid:
Risky, destination is the home directory
rsync -av data username@spmcluster:~/

If this happens and you still have an open session, you can fix it with:
chmod 700 ~

Always double-check the destination path before running rsync.